

Rajath Rao

Lead Software Engineer

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Software Engineer With 8 Years of Experience in Python, Cloud, and Backend Development.
Expertise in Cloud Architecture, Scalable Microservices, and GenAI Applications
Python, AWS Solutions Architect, Docker, Kubernetes, Terraform, FastAPI, Django, AI

EXPERIENCE

Intropic, UK - Lead Software Engineer

August 2025 - Present

- Design, build, and scale Python FastAPI microservices on AWS that power Intropic's AI-driven financial analytics platform, engineered for performance and reliability in production.
- Integrate generative AI into backend workflows through LLM orchestration, prompt engineering, RAG Workflows and cloud-native automation, turning financial data into AI-assisted analysis.
- Partner with data and research teams to take AI systems from prototype to production-ready services for AI-assisted decision-making.

TechM - Senior Software Engineer

Mar 2021 - Jan 2022

- Cyber Intelligence R&D at Tech Mahindra, in collaboration with Infoblox, building next-generation network-security solutions.
- Built FastAPI microservices for real-time monitoring and analysis of DNS server performance across enterprise networks.
- Designed and implemented CI/CD pipelines with application versioning to accelerate and streamline deployments. Embedded automated test suites into the pipeline, including BDD testing triggered on every commit to ensure continuous quality validation.

Goldman Sachs, UK - Vice President

July 2022 - Aug 2025

- Built and deployed a range of scalable Python applications and microservices on AWS - FastAPI, Flask, and PostgreSQL for mission-critical integration systems
- Led the upgrade to Python 3.11 and the migration from RDS to Aurora Serverless v2, improving latency, scalability, and cost-efficiency for high-demand applications
- Built and operated on-prem infrastructure for mission-critical integration systems, maintaining high stability and uptime for the IBD Engineering Loans division team.
- Engineered reusable infrastructure-as-code modules and database session-management components, accelerating delivery across the team.
- Refactored legacy systems to cut technical debt, reduce operational risk, and align with modern SDLC practices, and upgrading the existing CICD Pipelines on gitlab

NextOrbit - Software Engineer

Nov 2018 - Feb 2021

- Designed and implemented a custom reverse-index text-similarity algorithm powering the company's search and recommendation systems.
- Built data-ingestion pipelines for a data-lake architecture, enabling near-real-time ingestion and retrieval of multi-terabyte datasets.
- Partnered directly with clients to deliver business-intelligence reports in Tableau.

IBM, Python Data Analyst

Dec 2017 - Oct 2018

- Built an internal email-automation tool that cleared a chronic support-ticket backlog.
- Performed data cleaning and normalization in Python and identified automation opportunities across IT operations.
- Produced operational analytics on IT KPIs (SLA, incident, problem, change management) using Tableau and IBM Watson Analytics.

WEB FRAMEWORK

- FASTAPI
- DJANGO
- Flask
- DASH

PROJECTS

- **AWS** - Building a Web Application for an E-commerce Website
- **Django** - Personal Portfolio using Django
- **Kubernetes** - Chaos Monkey Implementation
- **NLP** - A Detailed Study of String Similarity Algorithms
- **AI** - Built a Neural Classifier to Predict Beer Ratings
- **Markov Decision** - eBay Car Sales Data, Understanding and predicting the sale of a particular car
- **Speech Recognition** - Simple Application to Convert Voice to Text Using AWS Transcribe

EDUCATION

Visvesvaraya Technological University

Bachelor of Engineering

Research Project

- Threat Detection in ATM Using Computer Vision
- Statistical Study of Treated Wastewater Using Nanotechnology

Personal Website

- [Portfolio Website](#)
- [Github](#)
- [Link For Projects](#)
- [Stack Overflow](#)

Coding Profiles

- [Leetcode](#)
- [Hackerrank](#)

CERTIFICATIONS

Certificate Verification Link

- AWS Solutions Architect
- AWS Cloud Practitioner - AWS CCP
- IBM Agile Thought Leader
- Data Structures and Algorithms
- AWS Developer Associate
- HashiCorp Terraform
- Claude Code Fundamentals
- GEN AI BootCamp

Programming Languages:

- Python
- JavaScript
- Golang
- TypeScript

Database Management

- RDBMS - MySQL, Postgres
- NoSQL - DynamoDB, MongoDB
- In-memory - Redis, Memcached
- Distributed - Cassandra

Tools

- Claude Code, Cursor, Codex
- Docker, Kubernetes
- Logstash, Kibana, Elasticsearch

GenAI / AI Stack

- LLM - OpenAI GPT, Claude, Llama
- APIs: OpenAI, Anthropic, Hugging Face
 - Agents: Tool calling, prompt chaining, multi-agent systems
- Claude Code, Codex, Cursor